Example 5.1

Find the general equation of a circle with centre $(-3,-4)$ and radius 3 units.

Solution

Equation of the circle in standard form is $(x-h)^2+(y-k)^2=r^2$

$$\Rightarrow \quad (x-(-3))^2+(y-(-4))^2 = 3^2$$
$$\Rightarrow \quad (x+3)^2+(y+4)^2 = 3^2$$
$$\Rightarrow \quad x^2+y^2+6x+8y+16 = 0.$$

Theorem 5.1

The circle passing through the points of intersection (real or imaginary) of the line $lx+my+n=0$ and the circle $x^2+y^2+2gx+2fy+c=0$ is the circle of the form $x^2+y^2+2gx+2fy+c+\lambda(lx+my+n)=0$, $\lambda \in \mathbb{R}^1$.

Proof

Let the circle be S : $x^2+y^2+2gx+2fy+c = 0$, ... (1)

and the line be L : $lx+my+n = 0$. ... (2)

Consider $S+\lambda L=0$. That is $x^2+y^2+2gx+2fy+c+\lambda(lx+my+n)=0$ ... (3)

Grouping the terms of x, y and constants, we get

$x^2+y^2+x(2g+\lambda l)+y(2f+\lambda m)+c+\lambda n=0$ which is a second degree equation in x and y with coefficients of x^2 and y^2 equal and there is no xy term.

If (α, β) is a point of intersection of S and L satisfying equation (1) and (2), then it satisfies equation (3).

Hence $S+\lambda L=0$ represents the required circle.

Example 5.2

Find the equation of the circle described on the chord $3x+y+5=0$ of the circle $x^2+y^2=16$ as diameter.

Solution

Equation of the circle passing through the points of intersection of the chord and circle by Theorem 5.1 is $x^2+y^2-16+\lambda(3x+y+5)=0$.

The chord $3x+y+5=0$ is a diameter of this circle if the centre $\left(\frac{-3\lambda}{2}, \frac{-\lambda}{2}\right)$ lies on the chord.

So, we have $3\left(\frac{-3\lambda}{2}\right)-\frac{\lambda}{2}+5 = 0$,

$$\Rightarrow \quad \frac{-9\lambda}{2}-\frac{\lambda}{2}+5 = 0,$$
$$\Rightarrow \quad -5\lambda+5 = 0,$$
$$\Rightarrow \quad \lambda = 1.$$

Therefore, the equation of the required circle is $x^2+y^2+3x+y-11=0$.

Example 5.3

Determine whether $x+y-1=0$ is the equation of a diameter of the circle $x^2+y^2-6x+4y+c=0$ for all possible values of c.

Solution

Centre of the circle is $(3,-2)$ which lies on $x+y-1=0$. So the line $x+y-1=0$ passes through the centre and therefore the line $x+y-1=0$ is a diameter of the circle for all possible values of c.

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